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Use Case 4 — MySQL Reporting & Integration

Objective: Store, query, and present analytical results from MySQL.

Tasks

1. Design & Populate Summary Tables

- Create tables and load the data (should be done thru python Mysql Connector).

```
# 1. Design & Populate Summary Tables
conn = sqlite3.connect(DB_PATH)
cursor = conn.cursor()
```

2. MySQL Queries

- Write SQL to compute:
  - Crime count per year
  - Top 5 crime types and their percentages
  - Arrest count per year.

```
# 2. SQLite Queries
# - Crime count per year
print("\n1. Crime count per year (from view):")
print(df_yearly[['Year', 'total_crimes']])

# - Top 5 crime types and their percentages
df_top5 = pd.read_sql("SELECT * FROM vw_crime_by_category LIMIT 5", conn)
print("\n2. Top 5 crime types and percentages:")
print(df_top5)

# - Arrest count per year
print("\n3. Arrest count per year (from view):")
print(df_yearly[['Year', 'total_arrests']])
```

3. Database Stored Views

- Create views in MySQL:
  - vw_crime_yearly
  - vw_crime_by_category

Yearly Crime Summary (Database View)

`SELECT * FROM vw_crime_yearly`

Year	Total Crimes	Total Arrests
2015	227	63
2016	242	62
2017	192	56
2018	213	72
2019	236	75
2020	224	68
2021	229	68
2022	221	72
2023	216	60

Top 5 Crime Categories

Most frequently reported crime categories

Primary Type	Crime Count	Percentage (%)
THEFT	416	20.8%
BATTERY	328	16.4%
ASSAULT	195	9.75%
NARCOTICS	178	8.9%
BURGLARY	177	8.85%

4. Pandas Integration

- Read these views into Pandas DataFrames for further analysis.

pd.read_sql("SELECT * FROM vw_crime_yearly", conn)

```
# 4. Pandas Integration
# Read these views into Pandas DataFrames for further analysis
df_yearly = pd.read_sql("SELECT * FROM vw_crime_yearly", conn)
df_category = pd.read_sql("SELECT * FROM vw_crime_by_category", conn)
```

5. Visualization from MySQL Data

- Plot SQL extracted data with Matplotlib and Seaborn.